

KSHITIZ BAKLIWAL

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🔗 [LeetCode](#) — [CodeChef](#) — [Codeforces](#) — [Codolio](#)

Education

Chandigarh University

Bachelor of Engineering in Computer Science and Engineering (CGPA: 7.55)

Aug 2022 – Jun 2026

Punjab, India

Technical Skills

Languages: Java, C++, TypeScript, JavaScript, SQL, HTML, CSS

Backend Frameworks: Spring Boot, Spring AI, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, Maven

AI & LLM Integration: Spring AI, OpenAI, Gemini, Ollama, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases, Semantic Search, Tool Calling, Model Context Protocol (MCP), Chat Memory, Structured Output, Streaming Responses, Multimodal AI

Frontend Development: React.js, Material UI

Databases & Cloud: MySQL, AWS

Developer Tools: Git, GitHub, Docker, Swagger, OpenAPI, Postman, Apache Tika, Apache PDFBox

Core Computer Science: Data Structures and Algorithms, Low-Level Design (LLD), Object-Oriented Programming (OOP), SOLID Principles, Multithreading, Exception Handling, Design Patterns (Strategy, Factory, Observer), AI Application Development

Projects

KnowFlow AI Platform | Spring Boot, Spring AI, OpenAI, Gemini, MySQL, Apache Tika, PDFBox **2025**

- Architected an AI knowledge platform using **Spring AI** with conversational chat, document intelligence, semantic search, and Multi-LLM integration for OpenAI, Gemini, and Ollama.
- Implemented **Retrieval-Augmented Generation (RAG)** using embeddings and vector search to enable contextual document Q&A, document comparison, translation, and automatic quiz generation.
- Engineered advanced AI features utilizing **Chat Memory**, **Streaming Responses**, **Structured Output**, and Tool Calling via prompt engineering techniques.
- Designed scalable REST APIs using a layered architecture, DTO patterns, input validation, Swagger OpenAPI documentation, and MySQL for persistent data storage.

Library Management System | Java, Spring Boot, Spring Security, React.js, MySQL, Docker **2025**

- Engineered a scalable full-stack web application featuring **JWT authentication** and granular role-based access control (RBAC) for Admin and Student modules.
- Designed REST APIs using **Spring Boot**, **Spring Data JPA**, and **Hibernate** with DTO architecture, global exception handling, pagination, and sorting.
- Built a responsive **React.js** user dashboard using Material UI and containerized backend services using **Docker** for consistent deployment.

Airbnb Backend Platform | Java, Spring Boot, React.js, MySQL, JWT **2025**

- Developed RESTful microservices handling property listings, room availability, booking pipelines, and automated customer management workflows.
- Configured **Spring Security** with stateless JWT authentication mechanisms to enforce strict role-based access control for Admins, Hosts, and Customers.

Low-Level Design (LLD) Systems | Java, Object-Oriented Programming, SOLID Principles **2025**

- **ATM Cash Dispenser System:** Built extensible architecture utilizing the **State Pattern** for machine states and **Strategy Pattern** for cash-dispensing algorithms.
- **Paytm Payment Gateway:** Modeled transactional entities leveraging the **Factory Pattern** for processing engines and **Observer Pattern** for real-time notifications.

Achievements & Problem Solving

- Solved **500+ Data Structures and Algorithms problems** across LeetCode, CodeChef, and Codeforces, focusing on Trees, Graphs, and Dynamic Programming.
- Secured **Global Rank 658** in CodeChef Starters 243 among thousands of competitive programmers.
- Maintained a **CodeChef 2-Star rating** (Highest Rating: 1450) with a dedicated Data Structures and Algorithms Rating of **1700**.

Certifications

- **NPTEL Cloud Computing** – Silver and Elite Certification awarded by IIT Kharagpur for demonstrating expertise in cloud infrastructure.
- **NPTEL Computer Organization and Architecture** – Bronze and Elite Certification awarded by IIT Guwahati for core computer hardware concepts.